

Thembinkosi Eden Thwala

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PROFILE

Software Engineer with 3+ years of experience designing, developing, integrating, testing, deploying, and supporting enterprise, cloud, and data-driven applications. Strong algorithmic and structured problem-solving ability, with practical experience translating business requirements into maintainable code, APIs, database workflows, asynchronous processes, and cloud-based solutions.

Proficient in Java, Python, C#, TypeScript, JavaScript, Ruby, and SQL, with experience across Angular, React, Ruby on Rails, ASP.NET *Core*, FastAPI, Flask, GraphQL, REST APIs, PostgreSQL, Redis, AWS, Azure, Docker, and ServiceNow.

Demonstrated ability to investigate previously unseen technical problems, learn unfamiliar technologies rapidly, communicate technical decisions clearly, and collaborate with developers, product owners, QA engineers, end users, and multidisciplinary teams. Applies system-level thinking to performance, scalability, maintainability, security, data consistency, failure handling, infrastructure requirements, and operational cost.

Committed to engineering standards, constructive technical criticism, clear documentation, reliable delivery, continuous learning, and building secure software solutions that integrate effectively with larger systems.

SKILLS SUMMARY

Programming Languages: Python, C#, TypeScript, JavaScript, Ruby, SQL

Algorithms & Problem-Solving: Algorithmic Thinking, Data Structures, Problem Decomposition, Logical Reasoning, Root-Cause Analysis, Performance Optimisation

Frontend Development: Angular, React.js, HTML5, CSS3, Tailwind CSS, Bootstrap

Backend Development: ASP.NET *Core*, Ruby on Rails, Node.js, FastAPI, Flask

APIs & System Integration: RESTful APIs, GraphQL, SOAP Services, Third-Party API Integration, Asynchronous Processing, Background Jobs

Databases: PostgreSQL, Microsoft SQL Server, MySQL, Oracle SQL, Redis, SQLite, Supabase

Cloud & Enterprise Platforms: AWS, Microsoft Azure, ServiceNow, Databricks, Docker, Kubernetes

Data Engineering & Analytics: Python, Pandas, NumPy, SQL, ETL Pipelines, Data Validation, Data Visualisation

Software Engineering: Object-Oriented Programming, Procedural Programming, Design Patterns, SDLC, Agile Scrum, System Design, Application Support, Technical Documentation

Testing & Reliability: Unit Testing, Integration Testing, API Testing, Debugging, Troubleshooting, Error Handling, Logging, Root-Cause Analysis

Security: JWT Authentication, OAuth 2.0, Role-Based Access Control, Authentication and Authorisation

DevOps & Version Control: Git, GitHub, GitHub Actions, CI/CD, Docker, Branching Strategies, Pull Requests, Code Reviews

Tools & IDEs: Visual Studio, Visual Studio Code, Postman, Jira, Confluence, Anaconda

EDUCATION

Diploma in Information Technology Software Development
Nelson Mandela University

- Graduated Cum Laude

2021/03 – 2023/12
Gqeberha, South Africa

Higher Certificate in Information Technology Support Services
Nelson Mandela University

- Graduated Cum Laude

2020/01 – 2020/12
George, South Africa

WORK EXPERIENCE

Software Engineer

Zensar Technologies

2025/07
Johannesburg,
South Africa

- Participating in enterprise technology training focused on Java, SQL, cloud technologies, software engineering best practices, and Agile methodologies.
- Developing Java applications and backend solutions while applying object-oriented programming principles and software design patterns.
- Building and consuming RESTful APIs and integrating applications with relational databases.
- Developing AI-powered chatbot solutions and intelligent automation prototypes using modern AI tools and APIs.
- Participating in internal innovation initiatives and hackathons, collaborating with multidisciplinary teams to design and deliver proof-of-concept solutions within tight deadlines.
- Working with Jira for sprint planning, backlog management, issue tracking, task estimation, and Agile project execution.
- Using Git and GitHub for version control, branching strategies, pull requests, and collaborative software development.
- Gaining practical experience with enterprise cloud platforms including AWS, Microsoft Azure, and ServiceNow.
- Performing debugging, troubleshooting, testing, and application support to improve software quality and reliability.
- Working with SQL databases to query, manipulate, and validate application data.

Junior Full Stack Developer

AgrigateOne

2024/02 – 2025/04
Johannesburg,
South Africa

- Designed, developed, tested, and maintained enterprise web applications using Ruby on Rails, Angular, TypeScript, Python, GraphQL, PostgreSQL, and Redis.
- Developed the Bulk Finance Rule Management module, implementing CSV import/export, validation, bulk create/update functionality, and automated error handling.
- Optimized PostgreSQL queries and backend services to improve application performance and scalability.
- Developed responsive user interfaces using Angular, Tailwind CSS, HTML5, and TypeScript.
- Worked extensively with AWS services including ECS, EC2, RDS, SES, SNS, and Textract to support cloud-hosted enterprise applications.
- Used Docker for containerized development and deployment across multiple environments.
- Developed and maintained asynchronous background jobs using Celery and Redis.
- Debugged production issues, investigated root causes, and implemented permanent software fixes.
- Collaborated with product owners, QA engineers, and developers using Agile Scrum methodologies, Jira, GitHub, and code reviews.
- Participated in sprint planning, backlog refinement, daily stand-ups, sprint reviews, and production deployments.

Graduate Cloud & DevOps Engineer

Disraptor

2022/11 – 2023/09
Johannesburg,
South Africa

- Supported cloud infrastructure hosted on Amazon Web Services (AWS) across development and testing environments.
- Provisioned, configured, and monitored AWS resources including EC2, RDS, IAM, S3, SES, Lambda, and CloudWatch.
- Assisted with deploying and maintaining cloud-native applications while following DevOps best practices.
- Automated deployment and operational tasks using scripting and cloud management tools.
- Configured IAM users, roles, and permissions following security best practices.
- Monitored application health, cloud resources, and system performance using AWS monitoring services.
- Assisted with troubleshooting cloud infrastructure, application deployments, networking, and production incidents.
- Supported CI/CD processes using Git, GitHub, Docker, and GitHub Actions.

PROJECTS

Thwala Attorneys Web Application [🔗](#)

2025/03 – 2025/06

Portfolio Project

Tech: React 18, TypeScript, Tailwind CSS, Python, FastAPI, Supabase PostgreSQL, Supabase Auth, SMTP Email

- Developed a full-stack legal services website and internal practice-management platform for Thwala Attorneys Inc.
- Built public contact, callback and consultation forms connected to a FastAPI backend and Supabase PostgreSQL database.
- Created separate staff and administrator dashboards for managing enquiries, consultations, appointments, notifications, team accounts and activity logs.
- Implemented role-based access control, automated enquiry workflows, threaded client communication, email notifications, document attachments and secure form validation.
- Deployed the frontend and backend on Render with production environment configuration and database migrations.

BudgetPro [🔗](#)

Portfolio Project

Tech: Next.js 15, TypeScript, Tailwind CSS, shadcn/ui, Supabase PostgreSQL, Supabase Auth, Row-Level Security, TanStack Query, Recharts, Brevo

- Developed a privacy-first budgeting platform that allows users to manage their finances without connecting or sharing online banking information.
- Built functionality for virtual accounts, income and expense transactions, transfers, custom categories, savings goals, debts and recurring payments.
- Implemented weekly, monthly and payday-to-payday budgets with overspending alerts, bill reminders and confirm-before-record recurring transactions.
- Created cash-flow forecasting, spending analytics, category reports, savings-rate calculations and interactive financial charts.
- Secured financial records using Supabase Authentication, PostgreSQL Row-Level Security and user-specific database access policies.

MinnieVerse [🔗](#)

Portfolio Project

Tech: Expo SDK 54, React Native, TypeScript, Expo Router, Python, FastAPI, asyncpg, Jinja2, Supabase PostgreSQL, Supabase Storage, Brevo

- Designed and developed a personalised mobile experience that transforms a child's tablet into an interactive themed charging and entertainment space.
- Built a React Native application that displays customised photos, videos, GIFs, affirmations, battery information and themed slideshow experiences.
- Developed a parent administration dashboard for managing themes, media, messages, connected devices and the active child experience remotely.
- Implemented secure device pairing, kiosk-style controls, charging-state monitoring, device heartbeats and remote content synchronisation.
- Created a FastAPI backend integrated with Supabase PostgreSQL, Supabase Storage and Brevo for data management, media storage and email verification.

Luna Rides [🔗](#)

Portfolio Project – Under Development

Tech: React 19, JavaScript, Tailwind CSS, React Router, Framer Motion, React Leaflet, Recharts, Python, FastAPI, MongoDB

- Designed and developed the frontend prototype for a premium, members-first e-hailing and chauffeur-booking platform.
- Created responsive interfaces for ride reservations, luxury vehicle categories, estimated fares, passenger requirements and scheduled trips.
- Prototyped customer experiences for airport transfers, corporate travel, group bookings, chauffeur matching and trip-status visibility.
- Integrated interactive maps, animated components, dashboards and data visualisations to demonstrate the proposed user experience.
- Structured the project for future FastAPI and MongoDB integration, with authentication, booking management, real-time trip tracking and payment functionality still under development.

Development Status: The deployed application currently demonstrates the frontend UI and intended user experience. Backend services, database integration and production workflows are still being developed. The final interface may change as development progresses.

ShuttleBee

Portfolio Project – Under Development

Tech: React 19, TypeScript, Tailwind CSS, React Router, TanStack Query, Framer Motion, Axios, Python, FastAPI, Pydantic, JWT, Supabase PostgreSQL

- Designed and developed the frontend prototype for a school transport management platform connecting parents, learners, drivers and transport providers.
- Created role-specific interfaces for managing learners, drivers, vehicles, routes, trips, subscriptions, announcements and support requests.
- Prototyped live route visibility, learner pickup and drop-off confirmations, trip updates, safety alerts and emergency-contact workflows.
- Designed operational dashboards that allow transport providers to coordinate fleets, monitor trips and manage daily school transport activities.
- Structured the platform for future FastAPI, JWT authentication and Supabase PostgreSQL integration, with backend workflows and real-time functionality still under development.

Development Status: The live deployment is currently a frontend UI demonstration.

Authentication, live location tracking, database operations, notifications and other backend functionality are still being developed. The final UI and platform workflows may change before production release.

Nkhensani

Portfolio Project – Under Development

Tech: React 19, JavaScript, Tailwind CSS, React Router, TanStack Query, Framer Motion, Python, FastAPI, Pydantic, MongoDB, Motor

- Designed and developed the frontend prototype for a premium hospitality platform that brings **FoodByNkhensani** and **CoffeeByNkhensani** together under one digital brand.
- Created dedicated pages for food services, coffee experiences, available services, brand information and customer bookings.
- Designed service experiences for private chefs, event catering, intimate dining, mobile coffee bars, corporate activations and combined hospitality packages.
- Built a responsive booking journey that allows customers to explore services, provide occasion details and review their selected experience.
- Structured the project for future FastAPI and MongoDB integration, including customer accounts, booking management, availability, payments and administrative functionality. The existing backend currently contains only an initial API and MongoDB foundation.

Development Status: The deployed application currently demonstrates the frontend UI, brand structure and proposed customer journey. The booking backend, database workflows, authentication, payments and administrative dashboard are still under development. The final interface and functionality may change before production release. The frontend currently uses React 19, React Router, Tailwind CSS and supporting UI libraries.

CERTIFICATES AND BADGES

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| • Databricks Certified Data Engineer Professional  | • Databricks Certified Data Analyst Associate  |
| • Cisco CCNA v7: Introduction to Networks  | • Cisco Introduction to Cybersecurity  |
| • W3Schools Certified Java Developer  | • Micro-Certification - Welcome to ServiceNow  |
| • Generative AI Application Deployment and Monitoring.  | • 120-Hour TEFL Certificate  |